Physicist-Engineer Nanotechnology and Quantum Applications Specialization Laboratory Measurement 1:

Qubit Manipulation in NV Diamond System

Abstract. Qubits are the fundamental building blocks of quantum technologies, enabling the coherent control and manipulation of quantum states for sensing and information processing applications. Among solid-state platforms, the nitrogen-vacancy (NV) center in diamond offers optical initialization and readout combined with long coherence times under ambient conditions.
In this measurement, you will encounter microwave-driven spin manipulation and optical detection techniques to investigate coherent dynamics, relaxation processes, and decoherence mechanisms. In particular, Rabi oscillations and relaxation times are measured to characterize the controllability and coherence properties of the system.

> **Warning.** This measurement makes use of a high power laser and a micorwave amplifier. These devices pose a significant safety risk if handled incorrectly. Carefully read the safety instructions found in Chapter IV **before** entering the lab and follow them at all times. Consult the lab instructor if in doubt and follow their instructions.

> **Warning.** It is recommended to bring your own laptop for this measurement. The required software is detailed in Prerequisites.

This development was carried out within the framework of the project 'Practice-Oriented Higher Education Infrastructure and Skills Development at BME' (Project ID: RRF-2.1.2-21-2022-00005), supported by a non-refundable grant from the Recovery and Resilience Facility funded by the European Union and the Government of Hungary.

The development of this lab exercise was funded by Faulhorn Zrt.

Introduction

I

1 Quantum bits as basis for future computing

The semiconductor industry has had a profound impact over the last several decades. Consequently, improvements in this field have been heavily pursued, leading to the formulation of the well-known Moore's law, which predicts a rapid increase in transistor density over time. However, in recent years the progress has begun to deviated from this expected trend, as modern devices approach the physical limitations of classical computation. The smallest commercially developed transistors are 2 nm wide. At this scale, classical physics can no longer describe the physical systems accurately. As a result, quantum mechanics must be incorporated into both the theoretical description and practical design of such systems. This naturally raises an interesting question: why should we stick with conventional transistor architectures? Why shouldn't we utilize the intrinsic quantum mechanical properties of nature to achieve fundamentally new computational capabilities?

The foundation of quantum computational systems are quantum bits or qubits, that are similar to classical bits in a sense that we require them to be in just two states when measured. But an important distinction between them is that while a classical bit can only be in either the 0 or the 1 state all the time, a qubit can also be in a coherent superposition of both. Operations on a qubit affect both states simultaneously while its in superposition. So, an operation on N qubits would extend on 2^N values, therefore exponentially increasing the parallelism and similarly decreasing the computational time. Important to note, that not every two state quantum system is suitable to be a qubit. There are certain conditions that need to be met in order to create a proper quantum computer. These are described by the DiVincenzo criteria [1].

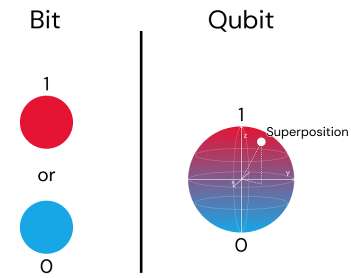


Figure I.1.1. Difference between classical and quantum bits.

Information may be physically realized in a variety of ways. One prominent platform is superconducting circuits, where microwave resonators and Josephson junctions are fabricated to form effective two-level systems. In trapped ions and neutral atoms, internal atomic states are utilized, while in photonic systems the qubit is typically encoded in the polarization of light. There are also options to implement a qubit in solid-state platforms. One such approach is based on semiconductor qubits, most commonly realized using quantum dots, whose quantum states are controlled by electrostatic gate voltages. Alternatively, qubits can be formed from crystal lattice defects in solids. These atomic-scale defects – also known as point defects – occur in wide-bandgap crystals, where the unique electronic structure of the localized defect states enables their use as qubits. Such point defects are the focus of this laboratory measurement. In particular, you will become acquainted with nitrogen-vacancy centers in a diamond environment.

2 Diamond embedded nitrogen-vacancy centers

2.1 Formation of NV centers

Diamonds are well known for being mechanically robust and having a very stable crystal structure. These properties make them desirable for a wide range of applications, but the high price of natural diamonds has limited their widespread use. Fortunately, artificial diamonds can now be produced easily and reliably, so much so, that these can be more pure – have less imperfections – than the naturally formed crystals. Because diamond growth techniques are now well-developed, it is possible not only to grow high-quality crystals but also to control what kind and how much imperfections can form in the crystal lattice.

During the growth process, nitrogen is introduced in the growth chambers which slowly builds in to the tetragonal carbon structure resulting in nitrogen point defects. Most commonly a nitrogen atom substitutes a carbon atom and bonds with the neighboring four carbon atoms, leaving one unpaired electron. To form nitrogen-vacancy (NV) centers, the crystal needs an additional type of defect: a vacancy, which is an empty lattice site. Vacancies are created by damaging the crystal in a controlled way, for example by exposing them to high-energy particles such as ion beams or neutron radiation from reactors. This step is followed by a heat treatment at around 800 °C. In the annealing process, the crystal heals itself and the vacancies become mobile, meaning they can move through the lattice. With a finite probability, a vacancy can reach a nitrogen atom and become trapped next to it, an NV center is formed. At the treatment temperatures, it is energetically favorable for the vacancy to remain beside the nitrogen rather than continue hopping around the lattice sites. This makes the NV center a stable defect, which can exist even at relatively high temperatures.

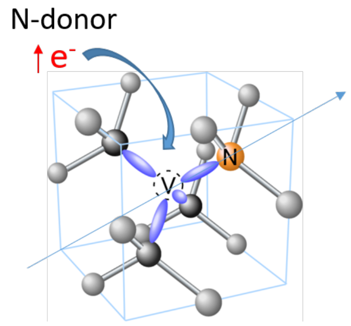


Figure I.2.1. Illustration of a negatively charged NV center [2].

Interestingly, this neutral NV center is not focus of most research. The charge-neutral state (NV^0) does not exhibit the unique properties that make it suitable as a qubit. What is usually referred to as the NV center is the one-times negatively charged state. The additional negative charge comes from the ionization of a substitutional nitrogen atom. The released electron becomes mobile and can be captured by the NV center, thereby forming the NV^- state. In most experimental conditions, the NV center is stabilized in this negatively charged state (NV^-), which is the charge state related to quantum control and optical readout. As this is the experimentally relevant configuration, any reference to an NV center in the following (and in general) will implicitly refer to the negatively charged state.

2.2 Energy states and spin polarization

The NV center effectively is a six-electron-system: 3 electrons come from the unpaired electrons of the three carbon atoms near the vacancy, 2 from the lone electron pair of the nitrogen atom and 1 from the ionized substitutional nitrogen. Due to symmetries of the crystal, they essentially create an electron system with a total spin number $S = 1$. So, the ground state of the NV center is a spin triplet with an energy level that would be three times degenerate. However, in electron systems with $S \geq 1$ arises an interaction between electrons called zero-field splitting. The result of this effect is that the states with different S_z components will split without the presence of outside magnetic field caused by the spin-orbit coupling and dipole-dipole interactions. In case of the NV center, the energy levels relating to the $|S_z = 0\rangle = |0\rangle$ and the $|S_z = \pm 1\rangle = |\pm 1\rangle$ states will separate with an energy difference of $D = 2.87\,\text{GHz}$.

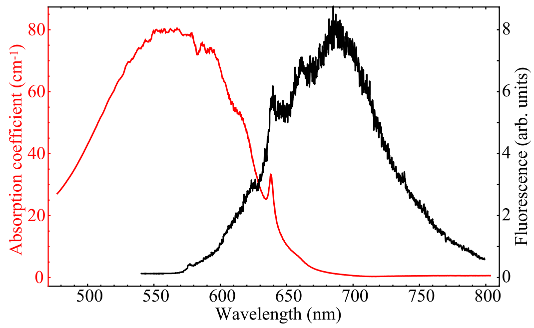


Figure I.2.2. Absorption and emission spectrum of the negatively charged nitrogen-vacancy center [3].

What makes the NV center unique is its optical cycle. From the spectra shown on Fig. I.2.2, it can be seen that the defect predominantly absorbs photons in the green region of the visible spectrum, while it emits mostly red light through a photoluminescence process. From an energy-level perspective, green light excites electrons from the ground state to a higher-lying excited state. From there, the electrons can relax back to the ground state via two different pathways: radiative decay (fluorescence) or a non-radiative intersystem crossing. These processes can be followed on Fig. I.2.3. An intersystem crossing corresponds to a transition from a triplet state to a singlet state. This singlet has only one possible spin projection, whose wavefunction is noted with $|0'\rangle$. The $|0\rangle$ triplet state can decay into the $|0'\rangle$ singlet state without changing its spin projection, whereas the $|\pm1\rangle$ states must have a spin transition as well, increasing or decreasing its S_z quantum number. This spin-changing transition is forbidden to first order, but there is a finite transition rate contributing to this event. In reality, the $|\pm1\rangle \rightarrow |0'\rangle$ transition has a higher probability than the $|0\rangle \rightarrow |0'\rangle$ one, leading to an effective depletion of the $|\pm1\rangle$ states. From the singlet state, a second intersystem crossing is required to return to the triplet ground state. This transition is also spin selective, the $|0'\rangle$ state is more likely to relax into the $|0\rangle$ ground state than into the $|\pm1\rangle$ states. Consequently, during an optical cycle a $|\pm1\rangle$ particle is likely converted into a $|0\rangle$ state. After many such cycles — corresponding to prolonged green-light excitation — the NV center is highly likely to be found in the $|0\rangle$ ground state. This process effectively polarizes the electron spin into a well-defined S_z sublevel.

Remark. The decay between singlet states may emit infrared radiation. This is filtered out in the setup.

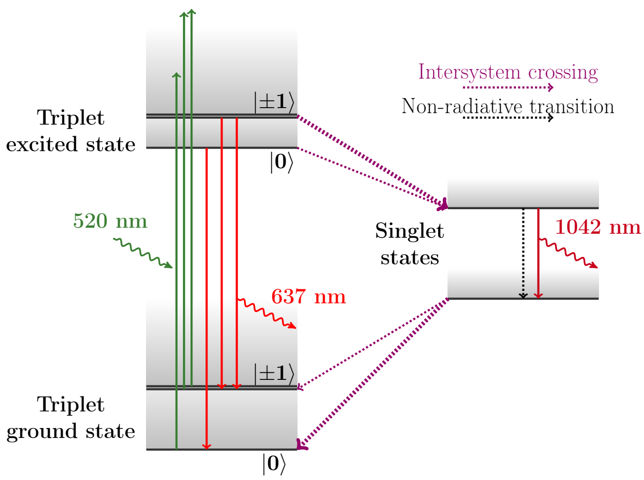


Figure I.2.3. Energy scheme of the negatively charged nitrogen-vacancy center [4].

The asymmetry of spin-sublevels is not only reflected in the intersystem crossing rates, but also in the fluorescence intensity. Since optical transitions are spin-state preserving, no spin flip occurs during the optical excitation and radiative decay. Because the excited $|\pm1\rangle$ states are more likely to decay into the singlet manifold, the probability of radiative decay — and thus the emission of a red photon — is reduced for these states. Essentially this means, that an NV center in the $|0\rangle$ state emit - on average - more photons than when it is in the $|\pm1\rangle$ states.

We conclude that we can initialize the spin-sublevel to the $|0\rangle$ state by extended laser excitation and read out the current spin-state by probing the fluorescence signal with a short laser pulse. High fluorescence indicates the $|0\rangle$ sub-state and low fluorescence indicates the $|\pm1\rangle$ sub-state.

3 Optically Detected Magnetic Resonance technique

While magnetic resonance is concerned with transitions between states that differ in their S_z spin quantum numbers, optical transitions occur between states that differ in their electronic configuration. Although magnetic resonance techniques have been well established for more than seventy years, their sensitivity is relatively low when compared to optical spectroscopic methods. In cases where the sensitivity of conventional magnetic resonance is insufficient, it is often possible to increase the population difference between the different magnetic sublevels by optical pumping. In optically detected magnetic resonance (ODMR) the aim is to combine magnetic resonance with laser spectroscopy in order to significantly improve sensitivity and to access additional information about the spin system. [5]

The central idea of ODMR is that magnetic resonance transitions are detected indirectly through changes in the optical signal. When a magnetic resonance condition is fulfilled, the redistribution of spin populations leads to a measurable change in photoluminescence intensity. This way, spin-state transitions can be observed with optical sensitivity.

Regarding the NV center, ODMR is realized by combining optical readout of the spin-state ($|0\rangle$ or $|\pm1\rangle$) with the energy splitting of spin-levels (zero-field splitting, Zeeman effect) in the gigahertz-range.

Measurement Methods

II

1 Theoretical Description

It is well known that measuring a quantum state without modifying it in the process is often impossible, the electron system of NV centers are no different. As described before, there is difference in photoluminescence intensity for photons emitted from different spin-sublevels, this means that we could potentially differentiate for which state the NV center is in. For one NV center (one physical qubit), one electron excitation would accompany one red photon emission, thus intensity difference would manifest as an average from multiple shots. As single photon detection goes beyond the scope of this lab course, the measurements will be done on a diamond sample that is dense with NV centers. This guarantees a large photon flux that can be measured with common photosensors, in hindsight, this NV center ensemble will only act as a pseudo-qubit.

An important part of qubits is the initialization, i.e. setting the initial state with the utmost precision, so that the time evolution of system is from a well-defined position. In NV qubits, the initial state is the $|0\rangle$ state. Unfortunately, both initialization and readout are performed by the same action: optical excitation. As such, we cannot read out the state without also pushing it towards the $|0\rangle$ state; meaning, a continuous readout will erase any information about the time-evolved state. This effectively eliminates the possibility of simply hooking the system up to an oscilloscope and measuring the transient signal.

1.1 Relaxation Time

In order to obtain information about the system, it must be probed for a controlled period of time; therefore, a pulsed laser scheme is required. The initialization pulse must be sufficiently long for the optical cycle to be repeated many times, ensuring that the NV centers are polarized into the $|0\rangle$ state. The readout pulse, on the other hand, should be short enough to minimize disturbance of the system, yet long enough to generate a sufficient number of photons to produce a measurable signal. Let us denote the time delay between the initialization and readout pulses by τ. As the delay is increased, the detected signal during the readout period decreases. This is due to relaxation processes that drive the polarized system back toward thermal equilibrium, where the electron populations are developed according to the Boltzmann distribution. Starting from the fully polarized state (with the NV centers predominantly in $|0\rangle$), thermal excitations gradually repopulate the $|\pm 1\rangle$ states. The decay – or relaxation – of the polarized state typically follows an exponential behavior with a characteristic time denoted by T_1. This is the timescale over which the spin system retains its polarization.

Remark. Larmor precession: An outside B_0 magnetic field causes the spin dipole moments to precess around the $\boldsymbol{B}_0$ field vector with an angular frequency of $\omega_{\text{Larmor}} = \gamma B_0$.

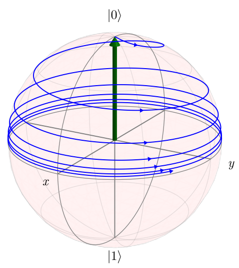


Figure II.1.1. Spin relaxation in the lab frame.

When a lock-in amplifier (see Lock-in Amplifier) is used, the T_1 measurement scheme is slightly modified (Fig II.1.2). Without a lock-in amplifier, the absolute fluorescence signal is detected. In contrast, a lock-in amplifier measures the difference (or contrast) between two alternating conditions. This is achieved by applying the readout pulse during one half of the lock-in reference period and omitting it during the other half. A contrast can only be meaningful, if the conditions are the same for both cases, i.e. there must be an initialization pulse at the beginning of both half-cycles. With this differential detection technique, only the signal modulation needs to match the timescale of the measured relaxation process, while the lock-in amplifier enhances sensitivity by rejecting uncorrelated background noise.

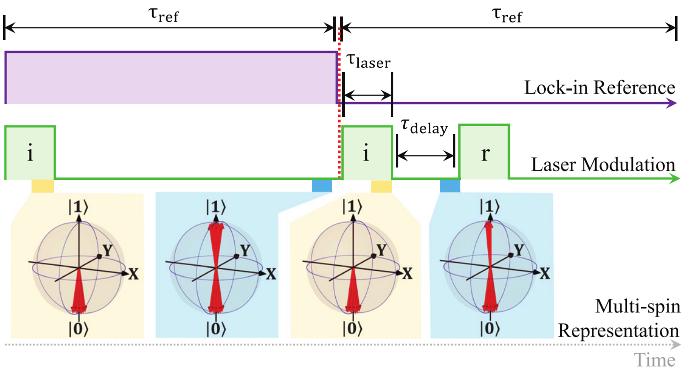


Figure II.1.2. Pulse sequence for measuring the T_1 relaxation time [6]. After the initializing laser pulses, the qubit states are in $|0\rangle$. Leaving the system freely evolve in time, the states relax back to their equilibrium state: (rougly) the same number of $|0\rangle$ and $|1\rangle$ states. However, if this propagation is interrupted with the readout pulse, then the fluorescence signal intensity coming from an intermediate state is related to its relaxation process.

1.2 Continuous Wave ODMR

You may have noticed that in the T_1 measurement no microwave (MW) field was applied. This is because we did not yet intentionally drive transitions between the spin states. Although the measurement still provides information about the spin system, it arises purely from the optical cycle, which polarizes into the $|0\rangle$ state. When a microwave field is applied, it can induce transitions between spin sublevels. In this sense, the MW can be regarded as a field that causes spin-flips, since absorption of a microwave photon changes the S_z spin projection by ± 1. This is demonstrated on Fig. II.1.4, the state vector is rotated from $|0\rangle$ into $|1\rangle$. So, the microwave acts as a driving force on the system: its phase determines the rotation axis, while its amplitude (or power) determines the rotation rate. The duration of the MW pulse controls the rotation angle, so spin-flip is not an instantaneous event, but rather a continuous transition between the two states.

The term Continuous Wave (CW) refers to the fact that the MW excitation is applied continuously rather than in pulses. In parallel, the laser is also kept on, providing constant optical excitation and spin polarization. In this situation, three processes compete with one another: the MW that would rotate the states; the optical pumping into $|0\rangle$ and the T_1 relaxation that drives the system toward equilibrium. Depending on the relative timescale of these processes, a steady state is formed in which the photoluminescence signal is constant. When the MW is turned off, the signal is maximal as the steady state becomes the $|0\rangle$ state. As the MW power (or in case of MW pulses, the pulse length) is increased, the $|1\rangle$ state gets more and more mixed into the NV centers' state. This results in the decrease of the photoluminescence signal due to the lower photon emission rate of the $|1\rangle$ state.

One essential property of the microwave field is its ω_{MW} angular frequency. Efficient rotation of spin states occur only under resonance conditions, when the microwave photon energy matches the energy difference between the spin sublevels. In the presence of an external magnetic field B_0, the Zeeman interaction contributes to the level splitting:

$$\Delta E = \hbar\omega_{\mathrm{MW}} = \gamma\hbar B_0,$$

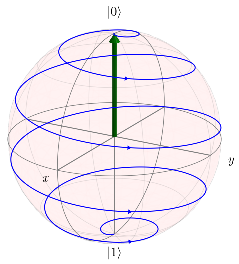


Figure II.1.3. Effect of a MW pulse in the lab frame.

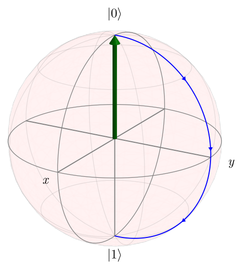


Figure II.1.4. Effect of a MW pulse in the rotating frame.

Remark. In lab frame, the disturbed spin rotates around the z-axis due to Larmor precession. This constant rotating movement can be transformed out, giving a simpler picture in the rotating frame.

where γ is the gyromagnetic ratio of electrons and $\hbar$ is the reduced Planck constant. Resonances may arise from the zero-field splitting, the Zeeman interaction, or hyperfine coupling. Whenever the resonance condition is satisfied, a dip appears in the fluorescence intensity. By sweeping the MW frequency, the resonant spin transitions can be mapped out. The transition lines are revealed as valleys.

When using a lock-in amplifier, the basic principle remains the same. Now the key information is from to MW resonance, thus the MW field needs to be modulated on and off during the two lock-in half period, while the laser remains continuously on. The lock-in amplifier then detects the fluorescence contrast between the MW-on and MW-off conditions, significantly improving sensitivity.

1.3 Rabi Oscillation

The Rabi cycle describes the cyclic evolution of a two-level quantum system in the presence of an oscillatory driving field. In the case of the NV center, a microwave field plays this exact role of the driving field. When applied on resonance, the MW field coherently rotates the NV spin state within the two-level subspace. The system is driven in and out of the pure basis states, and the populations oscillate harmonically between them. The inverse of one oscillation period defines the Rabi frequency, which scales linearly with the amplitude of the driving field: $\omega_{\text{Rabi}} = \gamma B_1$. To measure this effect experimentally, microwave pulses of varying duration are applied in order to see how long it takes for the system to rotate from $|0\rangle$ to $|1\rangle$. The resulting oscillatory fluorescence signal reveals the Rabi oscillations.

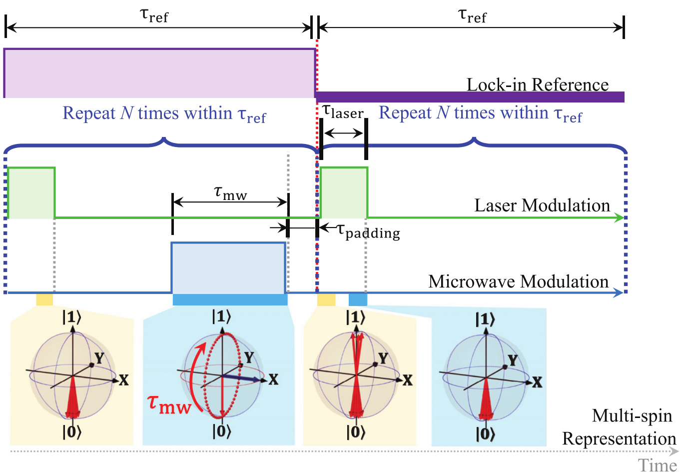


Figure II.1.5. Pulse sequence for observing Rabi oscillations [6]. The initializing pulses also serve as the readout and create the measured signal. The first pulse, without a preceding MW rotation, provides the baseline signal. In the second half-cycle, a MW pulse rotates the spin state before the laser pulse, the resulting change in the initial fluorescence yields the contrast. The disparity is enhanced when repeating the respective pulses in each half-cycles multiple times.

With help of the lock-in amplifier, the pulse sequence can be modified to directly measure contrast (Fig. II.1.5). Since lock-in detection requires two alternating conditions, one half of the reference period contains the MW pulse sequence, while the other half does not. In both half-cycles, an initialization pulse is applied to ensure identical starting conditions. However, a readout pulse is not strictly necessary as it turns out, it can be replaced by the second half's initialization pulse. You can look at this as if we were to shift the sequence in a way that the initialization pulses were at the end of the half periods. In other words, the readout effectively occurs at the beginning of the next initialization pulse, where the short-time fluorescence reflects the rotated spin state. The difference between the two half-cycles create the lock-in contrast and reveals the strength of the microwave-driven rotation. Since the T_1 relaxation typically takes place on a much longer timescale than the Rabi oscillations and because lock-in detection suppresses slow background variations, the MW and laser pulses can be repeated multiple times within one half-cycle. This significantly enhances the signal-to-noise ratio and reduces the total measurement time.

Although it may seem that the Rabi cycle can be measured for arbitrarily long MW pulse durations, it is not the case. If the time between two laser pulses becomes comparable to the T_1, relaxation processes begin to limit the measurable signal contrast. Moreover, additional decoherence mechanisms reduce the visibility of the oscillations. The spatial inhomogeneity of the driving MW field causes different NV centers to rotate at slightly different rates, conceptually resulting in incoherent rotations. Another important process is the transverse relaxation, or dephasing, characterized by the time constant T_2. This relaxation mechanism describes the loss of phase coherence in the transverse (x–y) plane and is directly connected to qubit information retention time. As a result, the Rabi oscillations are not perfectly sustained but exhibit an exponentially decaying envelope governed by the relevant (effective) coherence times.

2 Experimental Setup

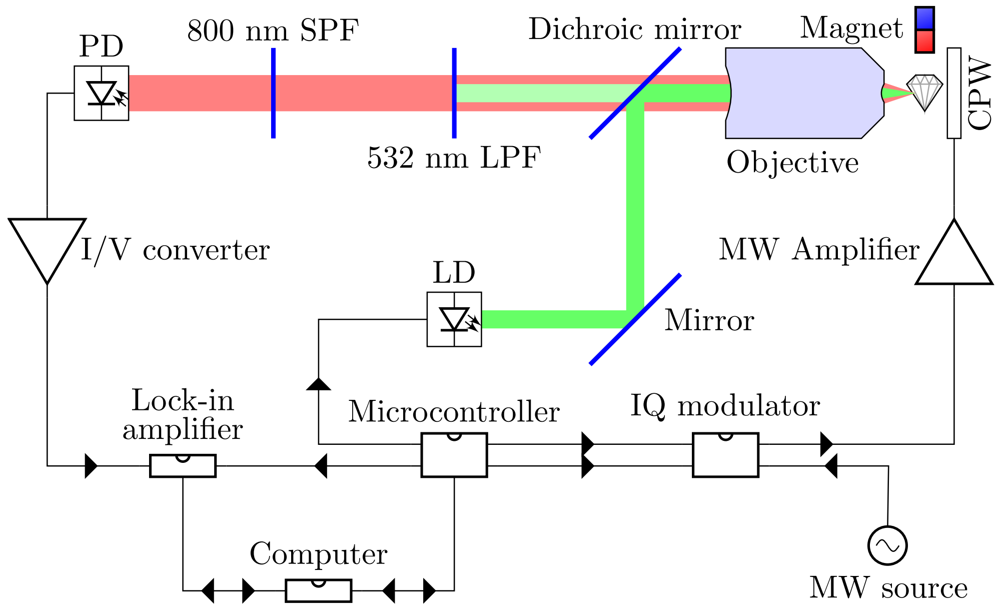


Figure II.2.1. Block diagram of the experimental setup.

2.1 Optics

Optical excitation is provided by a 520 nm green laser diode (ThorLabs L520P50), which is rated for a continuous output of 50 mW and can be modulated at up to 3 MHz by interrupting its power supply with a 2N7000 n-channel MOSFET (although higher frequencies are possible with a more sophisticated driver circuit). The diode's output is collimated and is reflected by an adjustable mirror (which serves to give extra degrees of freedom for alignment) onto a fixed dichroic mirror. This mirror element is highly reflective below a cutoff wavelength, while being transparent above it. The green light is focused onto the sample

Remark. The detector and the objective are mounted to an optical cage, so they do not need to be aligned manually.

by a microscope objective, and the produced photoluminescence is collected through the same objective as well. The collimated beam of red light passes through the dichroic mirror without reflection and is filtered to the desired wavelength range of 532 nm - 800 nm by a short and a long pass filter before hitting the detector.

2.2 Microwave Setup

The required microwave radiation is generated by a Kuhne MKU LO 8-13 PLL-2 oscillator, which has a variable output frequency. This signal is then fed into a Texas Instruments TRF370417EVM quadrature modulator (IQ modulator), which can do one of three things: either block the signal, let it through in-phase (I) or with a 90° phase shift (Q). This modulated signal then optionally goes through a Kuhne KU PA 270330-10A microwave amplifier, which can output up to 10 W power. Finally, the microwave enters an antenna called a coplanar waveguide. The sample is mounted on this waveguide. The microwaves that were not absorbed by the sample or radiated away are converted to heat by a 30 dB high power attenuator, while a $50\,\Omega$ termination eliminates reflections.

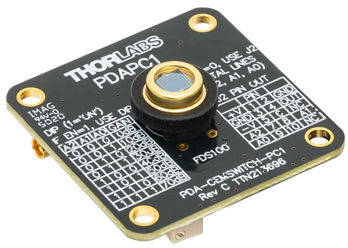


Figure II.2.2. PDAPC1 photodiode with integrated I/V converter.
Thorlabs, Inc.

2.3 Detection

The photoluminescence signal from the sample is detected by a Thorlabs PDAPC1 photodetector. This device includes a photodiode and the required I/V conversion circuits, all on a single PCB. The gain is adjustable, but changing it during the measurement will not be required. The detector's signal is connected to a Stanford Research Systems SR830M lock-in amplifier with an added $50\,\Omega$ termination.

2.4 Lock-in Amplifier

The measurement techniques (like the T_1 measurement) work fine on their own only if you have high-sensitivity sensors with low noise characteristics. Even if you have a state-of-the-art setup, getting as high of a signal-to-noise ratio is worth the while. That is why ODMR is usually complemented with a lock-in amplifier, in this setup we use a SR830M model from Stanford Research Systems. It acts as a band-pass filter, the decreased bandwidth also decreasing the shot noise. It can also extract small changes or transient signals that carry the relevant information of the fluorescence process. The first and second half of the lock-in period time naturally introduce a contrast feature between the two halves, as the first half is collected with a positive sign and the second half with a negative sign. The short photoluminescence process makes it possible to repeat the measurement frequently, utilizing that the amplifier integrates the incoming signal over a longer period of time (*conversion time*). Thus more measurements can be collected in one data point yielding a higher signal, while also decreasing its noise.

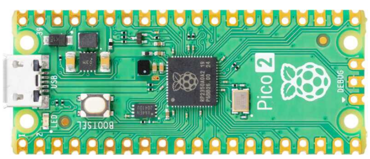


Figure II.2.3. Raspberry Pi Pico 2 microcontroller.
Raspberry Pi Ltd.

2.5 Computer Control

The control of the experimental setup is done via a Raspberry Pi Pico 2 microcontroller, which has been flashed with a custom firmware. This firmware allows it to synthesize precisely timed and synchronized pulse sequences on up to 5 channels. The device is used to generate a reference signal for the lock-in amplifier and for the modulation of the laser and the two microwave channels. The measurement control computer interfaces with the microcontroller and the lock-in amplifier to set up measurements and read the results.

Measurement Tools and Programming

III

List of Measurement Tools.

1. Custom measurement control board, including:
   - Pulse sequence generator (Raspberry Pi Pico 2 with pico-pulse firmware)
   - Quadrature modulator (Texas Instruments TRF370417EVM)
   - Laser power supply and modulator
2. Lock-in amplifier (Stanford Research Systems SR830M)
3. Photodiode with I/V converter (Thorlabs PDAPC1)
4. Lab bench power supply
5. Microwave assembly:
   - Microwave oscillator (KUHNE MKU LO 8-13 PLL-2, Oscillator)
   - Microwave amplifier (KUHNE KU PA 270330-1)

A python library has been created to help interface with the measurement devices. This library is available on GitHub. The Devices subdirectory contains the relevant device drivers and working experimental code is found in Experiments. While the latter is available for use should it be necessary, students are expected to write their own code, as it will contribute to the lab notes score.

1 Prerequisites

To be able to interface with the measurement setup using the provided drivers, make sure your laptop has:

- A copy of the driver code. Clone the GitHub repository locally or download the zip file from the website.
- A Python environment of your choice. Jupyter notebooks are recommended, as you can write your lab notes in them as well.
- A VISA driver, either NI-VISA or the `pyvisa-py` package
- The following python packages (also included in a requirements.txt file in the provided code):
  - `pyvisa` – interfacing with instruments
  - `pyvisa-py` – standalone VISA driver, strongly recommended for Linux
  - `numpy` – math

- pandas – tabular data
- pyserial – serial communication
- matplotlib – plotting
- tqdm – progress bars

2 Pulse Sequence Generator

The heart of the setup is the pulse generator. It is responsible for generating the modulation signal for the laser and microwave excitation, as well as providing the reference clock for the lock-in amplifier. This enables the use of stroboscopic techniques and makes the time-resolved probing of the system possible.

The device itself is a cheap, off-the-shelf Raspberry Pi Pico 2 microcontroller. The custom firmware makes use of the PIO (programmable input/output) coprocessors of the RP2350 chip (for which the Pico 2 serves as a carrier board) to generate synchronized pulses on up to 5 channels with a precision of 1 CPU cycle. In case of the slightly overclocked devices included in the setup, this corresponds to a temporal resolution of 5 ns. Due to how the PIO units obtain data and generate the pulses, the shortest pulse is 4 cycles long. Note that the device will automatically round everything up to 4 and down to the nearest integer number of cycles, but does so silently. Make sure you do not send invalid durations, as this will cause a discrepancy between what you requested and the the impulse sequence that was actually generated.

The `PicoPulse` class is provided in `Device.PicoPulse` for ease of use. Most importantly, the `sendSequence` method takes a pandas `DataFrame` with columns `time` and `ch1` to `ch5`, decodes it and uploads it to the device. If no keyword arguments are supplied, the time is interpreted in nanoseconds and the sequence will be repeated indefinitely. On the control board, `ch1` corresponds to the lock-in reference, `ch2` to the quadrature and `ch3` to in-phase channel of the modulator, and `ch4` to the laser's modulation (`ch5` is unused). If you pass a pin definition dictionary to the PicoPulse constructor you can also use the defined names instead of the channels. For example, to produce a 1 MHz square wave on the lock-in output (`ch1`) and a 2 MHz square wave on the laser output (`ch4`), use the following code:

```python
import pandas as pd
import pyvisa

from Devices.PicoPulse import PicoPulse

pico_pins = {
    'lockin': 'ch1',
    'Q': 'ch2',
    'I': 'ch3',
    'laser': 'ch4'
}

seq = pd.DataFrame(
    columns = ['time', 'lockin', 'laser'],
    data = [
        [250, 0, 0],
        [250, 0, 1],
```

```
            [250, 1, 0],
            [250, 1, 1]
        ]
    )

    rm = pyvisa.ResourceManager()
    pico = PicoPulse(rm, 'ASRL3::INSTR', pico_pins)
    pico.sendSequence(seq)
    rm.close()
```

To help with constructing pulse sequences, `Utilities.SequenceVisualizer` contains two helpful function. `visSeqProportional(seq)` draws the sequence as a function of time, while `visSeqEquidistant(seq)` will stretch everything to be the same width, but add duration labels.

3 Lock-in Amplifier

The `SR830M` class from `Devices.LockIn` provides functions for interfacing with the lock-in amplifier at a high level. Notably, the `snapshot` method takes a list of up to 6 strings with the desired channel names and returns the aptured values. The `multiRead` method helps capture a time series using the internal sampling capabilities of the lock-in amplifier. This is especially useful for taking multiple readings from the same experiment in order to take an average and to calculate noise. Note that the device can only store the currently displayed metric, so exactly two channels are available with the options listed under the corresponding display (i.e. X-Y is possible, but X-R isn't, as they're on the same display). Most front panel settings also have corresponding methods, exhaustive documentation of which can be found in the source file. The following code demonstrates how to read data from the device using the library:

```
from Devices.LockIn import SR830M

rm = pyvisa.ResourceManager()
lockin = SR830M(rm, 'GPIB0::8::INSTR')

# Read single values
x, y, r, theta = lockin.snapshot(['x', 'y', 'r', 'theta'])
reference, aux1 = lockin.snapshot(['ref', 'aux1'])

# Sample X and Y for 8 seconds with an automatic sample rate
# (calculated from time constant)
xs, ys = lockin.multiRead('x', 'y', 8)

# Sample AUX4 for 10 seconds at 4 Hz
# Setting channel 1 to None speeds up readout
_, aux4s = lockin.multiRead(None, 'aux4', 10, 4)

rm.close()
```

4 Microwave Source

The device class `KuhnePLL` is included in `Devices.LO`, with the only notable method being `setGHz(val)`. The microwave cannot be disabled from the oscillator itself, and as such the quadrature modulator must be used to block it if you do not wish to use it. This device uses raw serial communication instead of VISA for communication, so the destructor must be called properly after the device is no longer needed. Instantiating the device without first destructing the previous instance will result in a permission error, which can be cleared by power cycling the device. A working example can be seen below:

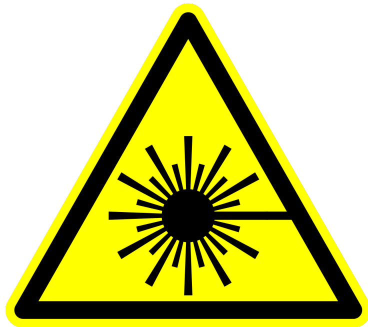
```python
from device.LO import KuhnePLL

osc = KuhnePLL('COM4')
osc.setGHz(2.87)

del osc
# or
osc = None
```

IV

Measurement Tasks



Warning. This experiment involves the use of a class IIIb laser diode with 100 mW peak power. Direct or reflected beams will cause **permanent eye damage**. Diffuse reflections can cause pain and should only be viewed for short periods of time.

Mandatory safety precautions:

- Do not look into the direct or reflected beam.
- Remove reflective accessories (e.g. rings, bracelets and watches) before adjusting the setup.
- Keep your head above the optical plane at all times.
- Do not adjust the laser's power supply. Laser diodes are nonlinear components, so even a small increase in voltage may result in dangerously high brightness and a burnt out diode.
- If unsure, consult the lab supervisor.

Warning. This experiment uses high power microwave radiation (up to 10 W). Improper use may result in bodily harm and the destruction of equipment.

Mandatory safety precautions:

- Microwaves should always be absorbed by 50 Ω terminations. Standard terminations are sufficient on their own for low powers, but use them in conjunction with a high power attenuator for anything after the amplifier.
- Always unpower the amplifier before modifying the connections. High power reflections will damage the amplifier.
- Avoid exposure to microwave radiation. Do not touch the coplanar waveguide during operation.

Warning. The photodiode's output must be terminated with 50 Ω. The lock-in amplifier has a high input impedance, so a parallel termination is required. Remove the parallel termination when connecting the photodiode to a device which already has a 50 Ω input impedance, as a double-termination may result in unacceptably high current through the system.

1 Preparation

Task 1. Assemble the setup.

Connect the laser to the control board using the 2 pole connector. Attach the BNC output of the control board to the reference input of the lock-in amplifier. Connect the photodiode to the lock-in's input with a BNC cable. Power up the control board by plugging a 9V power supply into the barrel jack.

Set the lab power supply to 12 V and 0.15 A on both channels. Connect the positive output of CH1 and the negative output of CH2 to the ground potential using the included cable. The green banana jack should be connected to the ground potential, the black one to the negative output of CH1 (-12 V) and the red one to the positive output of CH2 (12 V). **Ask for the supervisor's approval before enabling the outputs!**

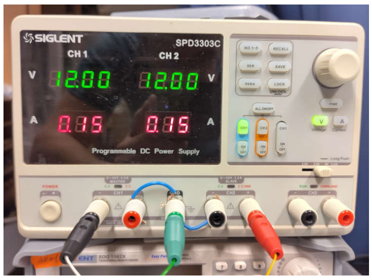


Figure IV.1.1. Correct settings and wiring for the photodiode's power supply. Your exact power supply may be different, but the same principle applies in all cases.

Connect the AUX output of the oscillator (found on the microwave assembly) to the one of the differential LO inputs of the quadrature modulator, while terminating the other one with 50 Ω. For now, connect the output of the quadrature modulator directly to the coplanar waveguide, skipping the amplifier (for CW measurements a lower microwave power is desired). Make sure the other port of the waveguide is connected to a high power attenuator and then a 50 Ω termination.

Task 2. Adjust the setup to maximize the photoluminescence signal.

Use one of the Aux In ports of the lock-in amplifier as a voltmeter. The range is 0-10 V by default, but the limit can be lowered using the "expand" option under the display.

Hint for the Lab Notes. Make sure to note your achieved signal levels. This will be useful later on as a point of reference.

2 Continuous Wave Experiments

Task 3. Implement CW sequence and optimize for lock-in signal.

Remark. Feel free to ask the lab instructor for hints on how to construct the measurement code. If needed, the reference code is available for use, although this will reduce the maximum score you can achieve on the lab notes, as the code is taken into account when grading.

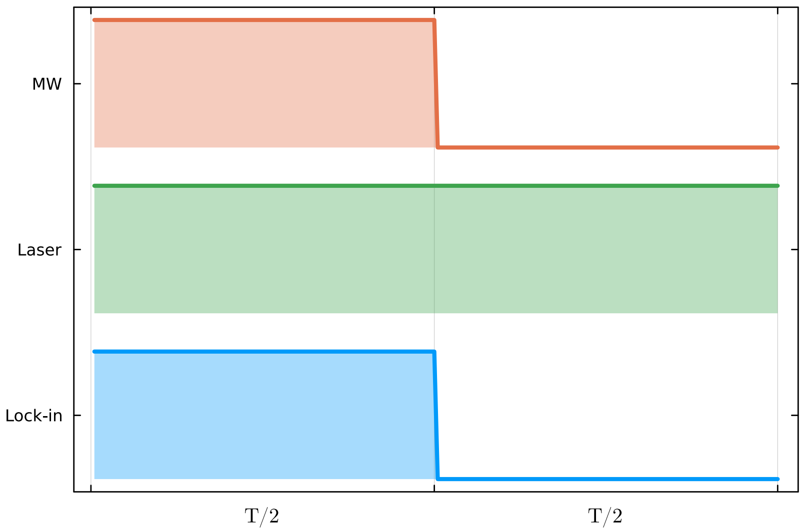


Figure IV.2.1. CW pulse sequence scheme.

Hint for the Lab Notes. Compare your ODMR signal level to the photoluminescence signal.

Hint for the Lab Report. Did you need to to move the focus spot on the diamond to get a better signal? If yes, why are the photoluminescence and ODMR signal maxima at different locations?

Task 4. Sweep the microwave spectrum and record the ODMR signal in zero magnetic field.

Using the previously constructed sequence, repeat the measurement with different microwave frequencies. Measuring between 2.85 GHz and 2.9 GHz is a good start. Refine the range and produce a spectrum with at least 100 points.

Hint for the Lab Notes. How many peaks do you see? Which transitions do they correspond to?

Hint for the Lab Report. Extract the zero filed splitting parameters D and E. D corresponds to the average frequency, while E is half the distance between the peaks.

Task 5. Sweep the microwave spectrum and record the ODMR signal with an arbitrary magnetic field.

Repeat the previous measurement with a magnet placed near the sample. For optimal results, ensure that the magnetic field is not parallel to any of the possible NV center orientations (the face of the sample is normal to the [111] direction). Note that you will need to greatly increase your frequency range; 2.6 GHz to 3.1 GHz is a good start.

Hint for the Lab Notes. How many peaks do you see? Note that there are 4 possible orientations for an NV center in the lattice, and they are all present in equal numbers in most samples.

Hint for the Lab Report. Explain why you see the number of peaks that you do.

Remark. The spectrum may be a bit “wobbly”, this is expected and completely fine. The microwave oscillator in the setup was designed to power a radio beacon, not to serve as a scientific instrument, and as such the output power is not stabilized. As the frequency is varied the power will fluctuate by an order of magnitude. This is somewhat compensated by the quadrature modulator, but is still very noticeable in the resulting spectrum.

Task 6. Observe hyperfine splitting.

Choose a prominent peak and focus on it to record a high resolution spectrum. With low microwave power (ask the supervisor for attenuators), you should see that the resonance peak is actually split into three due to the hyperfine interaction with the neighboring nitrogen atom.

3 T_1 Relaxation

Task 7. Measure T_1 relaxation.

Implement the pulse sequence (see Pulse Sequence Generator) and measure the signal level for at least 10 values of τ. We're only interested in the absolute change in signal caused by the readout pulse, so it is recommended to use the $R - \Theta$ mode of the lock-in. Choose the value of t_{pad} such that $\tau \ll t_{\text{pad}}$ and $\tau + t_{\text{pad}} = \text{const.}$ for all values of τ.

Remark. The timescales vary by sample. Consult the lab supervisor for a good range for τ.

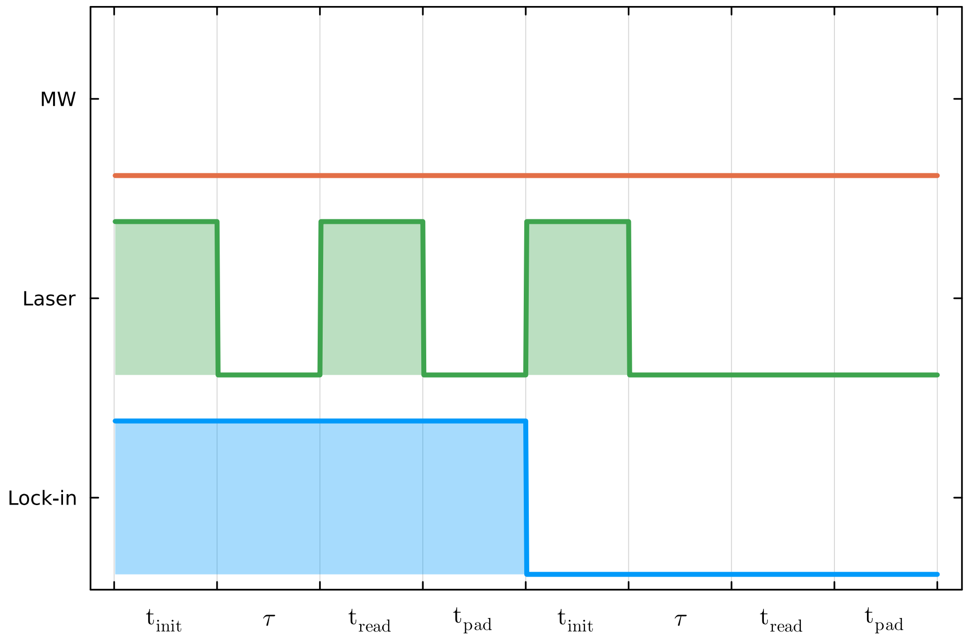


Figure IV.3.1. T_1 pulse sequence scheme.

Hint for the Lab Notes. Note how long your initialization and readout pulses were. Plot the signal level as a function of τ.

Hint for the Lab Report. Recover the T_1 characteristic time by fitting the signal with the following model:

$$V_R(\tau) = Ae^{-\tau/T_1} + C \tag{IV.3.1}$$

Hint for the Lab Report. Determine the contrast between the signal level of the polarized and relaxed states. For reference, the best contrast achieved on the prototype measurement setup corresponded to a 13 % increase in lock-in signal when the system was in an initialized state compared to the equilibrium.

4 Rabi Oscillation

Task 8. Measure Rabi oscillations.

Warning. This measurement makes use of a microwave amplifier. This device can produce up to 10 W of radiation and should be handled accordingly. Use attenuators **before** the amplifier, as only certain special attenuators are rated for such high power.

Implement the pulse sequence and observe Rabi oscillations. Choose the value of t_{pad} such that $\tau + t_{\text{pad}} \ll T_1 \approx 1$ ms and $\tau + t_{\text{pad}} = \text{const.}$ for all values of τ.
Connect the amplifier between the output of the quadrature modulator and the coplanar waveguide to produce a higher B_1 field and thus faster oscillations. With the external magnetic field still present, tune into a prominent resonance peak. Vary the microwave excitation time between 20 ns and 5 μs and record the corresponding signal levels.
Functions are provided in the chapter Lock-in Amplifier to automatically perform repeated measurements on the lock-in. The resulting data series can be used to calculate the mean signal and its standard error.
If time permits, insert attenuators between the modulator and amplifier to reduce output power and repeat the measurement.

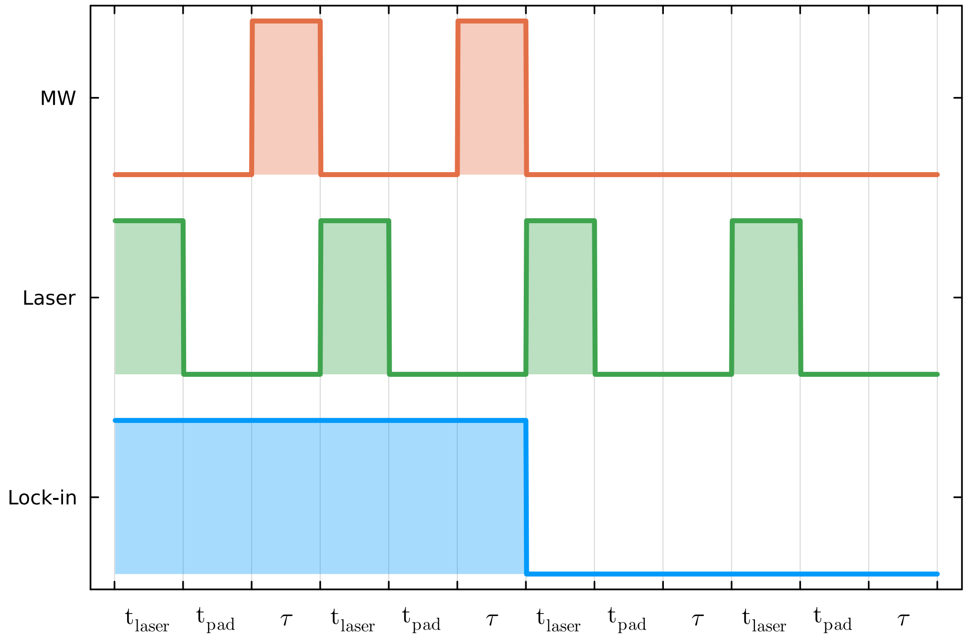


Figure IV.4.1. Rabi pulse sequence scheme. When varying τ, make sure the total length of the sequence remains constant by also adjusting t_{pad}. Each half of the sequence has been repeated twice here, but it's recommended to repeat it 10-100 times for best effect.

Hint for the Lab Notes. Plot the signal as a function of microwave excitation time. What do you see? If you repeated the measurement with different powers, what did it change?

Hint for the Lab Report. Extract the Rabi frequency from the measurements. Experience shows that the textbook formula is a bad fit, so it needs to be modified.

A linear term is added to account for microwave heating. The exponential envelope is replaced with a stretched exponential with empirical parameter β to account for inhomogeneity.

$$V_R(\tau) = Ae^{-\left(\tau/T_2^{\mathrm{eff}}\right)^\beta} \sin\left(\omega_{\mathrm{Rabi}}\tau + \phi\right) + B\tau + C \tag{IV.4.1}$$

Hint for the Lab Report. If you repeated the measurement with different microwave powers, compute the Rabi frequency for all measurements. Is there a correlation between power and frequency? Does this support the claim that you were measuring Rabi oscillations and not some unrelated phenomenon?

Contribution

The measurement setup was developed by Bence Göblyös based on a setup described in literature [6] under the supervison of Prof. Ferenc Simon and Robin Kucsera. The lab instructions were written by Robin Kucsera and Bence Göblyös. Samples were provided by Dr. Sándor Kollarics.

This lab exercise was developed with funding and consultation from Faulhorn Zrt.

Please contact Bence Göblyös at bence@goblyos.dev if you have any questions regarding the experimental setup or the driver code.

References

[1] David P DiVincenzo. Topics in quantum computers. *NATO ASI Series E Applied Sciences-Advanced Study Institute*, 345:657, 1997.

[2] A. Gali. Ab initio theory of the nitrogen-vacancy center in diamond. *Nanophotonics*, 8(11):1907–1943, 2019.

[3] Victor Marcel Acosta. *Optical magnetometry with nitrogen-vacancy centers in diamond*. Phd thesis, University of California, Berkeley, 2011.

[4] Sándor Kollarics. *Magneto-optical spectroscopy on qubit candidate solid state systems*. PhD thesis, Budapesti Műszaki és Gazdaságtudományi Egyetem, 2023.

[5] Dieter Suter. Optical detection of magnetic resonance. *Magnetic Resonance Discussions*, 2020:1–47, 2020.

[6] Vikas K. Sewani, Hyma H. Vallabhapurapu, Yang Yang, Hannes R. Firgau, Chris Adambukulam, Brett C. Johnson, Jarryd J. Pla, and Arne Laucht. Coherent control of nv-centers in diamond in a quantum teaching lab. *American Journal of Physics*, 88(12):1156–1169, 12 2020.